In many sectors in Mathematics, the 'Invariant' objects are quite important.

[Def. Let K/F be a finite separable extension, and let L/F be Galois closure. Define

$$N_{K/F}: K \to F; \alpha \mapsto \prod_{\sigma \in Emb_F(K,L)} \sigma(\alpha)$$

$$Tr_{K/F}: K \to F; \alpha \mapsto \sum_{\sigma \in Emb_F(K,L)} \sigma(\alpha).$$

In particular, if K/F is Galois, then $L=K$, so that $Emb_F(K/L) = Gal(K/F)$.

[Prop.

1). $N_{K/F}$ and $Tr_{K/F}$ are well-defined. (That is, for any $\alpha \in K$, $N_{K/F}(\alpha), Tr_{K/F}(\alpha) \in F$)

2). The $N_{K/F}$ preserves multiplication and the $Tr_{K/F}$ preserves addition.

Proof. Let